Lab 6: Code and Branch Coverage Analysis

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1. Result

Test file	Tests	Line coverage	Branch coverage	Evidence
CodeCoverageTest_income_tax.py	16	100% (39/39 statements)	99% (27 of 28 branches; missing 7->9)	code_coverage.png
BranchCoverageTest_income_tax.py	33	100% (39/39 statements)	100% (28 of 28 branches)	branch_coverage.png

Both are above the 90% target. Commands: `coverage run -m pytest CodeCoverageTest_income_tax.py` and `coverage run --branch -m pytest BranchCoverageTest_income_tax.py`, each followed by `coverage report -m --include=income_tax.py` (the report is limited to the application file so the test file does not inflate the total).

2. Code coverage analysis

The starter file had two tests. I added 14 more so that every statement runs: the invalid-income and unknown-regime errors, the old-regime investment cap, income below the deduction, all three age groups, both tax bands, the invalid-amount error, and each of the five `refund_status` results. This gives 100% line coverage (39 of 39 statements).

3. Branch coverage analysis

Line coverage said 100%, but the branch report for the same tests showed **99% with one partial branch, 7->9**. Line 7 is `if investments > 150000:`. Every old-regime test I had used investments above 150000, so the condition was never False. The line ran, but the 'skip the cap' path (7 to 9) never did. Line coverage hides this, because the only statement inside the if is already counted as executed.

Branches in the code (28 in total):

Function	Decisions	Branches
taxable_income	negative income, regime new / old / unknown, investments over the cap, taxable below zero	10
tax_due	negative amount, age 80+, age 60+, taxable within exemption, slab within 250000	10
refund_status	difference above zero, on time (refund side), difference below zero, on time (balance side)	8

Tests added in BranchCoverageTest_income_tax.py (17 more, 33 in total):

Group	Tests	What they check
taxable_income	5	investments below the cap (closes branch 7->9), investments exactly 150000, old-regime income below deduction, income 0, income equal to the deduction
tax_due boundaries	10	taxable 0; exactly at the exemption limit (250000); just above it (250001 gives 0 because of integer division); ages 59, 60, 79 and 80; slab exactly 250000 and just above; upper band for seniors and super seniors
refund_status	2	zero paid and zero due; settled does not depend on the filing date

4. Conclusion

The first test file reached 100% line coverage and still left one decision untested. Adding boundary and both-outcome cases took branch coverage from 99% to 100%. High coverage shows that code was run, not that its results are right, so each test also asserts the exact expected value (for example, tax_due(800000, 30) = 72500).